

Optimization

Minimization and Karnaugh Maps

- You can simplify, reduce the cost of, the circuit (i.e. logical expression) using two ways:

1. Boolean Algebra properties
2. K-map

- A Karnaugh Map (K-Map) is a visual method used in digital system design to simplify Boolean expressions, reducing hardware complexity and cost.

- It organizes truth table values into a grid.

- A grid representation of the truth table, where adjacent cells differ by only one variable.

- Simplify (or minimize) SOP functions (circuits):

1- Two Variables Map:

$$f(x_1, x_2) = \bar{x}_1 \bar{x}_2 + \bar{x}_1 x_2 + x_1 x_2$$

x_1	x_2	f
0	0	m_0 1
0	1	m_1 1
1	0	m_2 0
1	1	m_3 1

		x_1	
x_2	0	1	
	0	1	
	m_0	m_2	
	1	1	
	m_1	m_3	

		x_2	
x_1	0	1	
	0	1	
	m_0	m_1	
	1	1	
	m_2	m_3	

2- Three Variables Map:

x_1	x_2	x_3	
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

		$x_1 x_2$			
		00	01	11	10
x_3	0	m_0	m_2	m_6	m_4
	1	m_1	m_3	m_7	m_5

		$x_2 x_3$			
		00	01	11	10
x_1	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

Example 1: Simplify the following using a K-map

$$f(x_1, x_2, x_3) = \bar{x}_1 \bar{x}_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 \bar{x}_3$$

x_1	x_2	x_3	
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

		00	01	11	10
	0				
	1				

Example 2: Give the minimum cost representation of the following logical expression

$$f(a,b,c) = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b} + ab\bar{c}$$

x_1	x_2	x_3
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

	00	01	11	10
0				
1				

Multiple examples:

	00	01	11	10
0				
1				

	00	01	11	10
0				
1				

	00	01	11	10
0				
1				

3- Four Variables Map:

X1	X2	X3	X4	f
0	0	0	0	m_0
				m_1
				m_2
				m_3
				m_4
				m_5
				m_6
				m_7
				m_8
				m_9
				m_{10}
				m_{11}
				m_{12}
				m_{13}
				m_{14}
				m_{15}

x_3x_4 \ x_1x_2		x_1			
		00	01	11	10
00	m_0	m_4	m_{12}	m_8	
01	m_1	m_5	m_{13}	m_9	
11	m_3	m_7	m_{15}	m_{11}	
10	m_2	m_6	m_{14}	m_{10}	
		x_2			
		x_4			

$x_3 x_4$ \ $x_1 x_2$		00	01	11	10
		0	1	3	2
00		4	5	7	6
01					
11		12	13	15	14
10		8	9	11	10

Example 1: Find the minimum cost SOP expression for the following function. (or Design the simplest SOP expression that implements the following expression)

$$f(x_1, x_2, x_3, x_4) = \sum m(2, 3, 9, 10, 11, 13)$$

x1	x2	x3	x4	f
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

	00	01	11	10
00				
01				
11				
10				

$x_1 x_2$ $x_3 x_4$	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	1	1	0	1
10	1	1	0	1

	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	1	1	1	1
10	1	1	1	1

	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	1	1	1	0
10	1	1	0	1

	00	01	11	10
00				
01				
11				
10				